Hand-Gesture Controlled
Pick & Place Robot

Robotic Arm Simulation | Control & Integration Project

Academic Project · Robotics & Automation



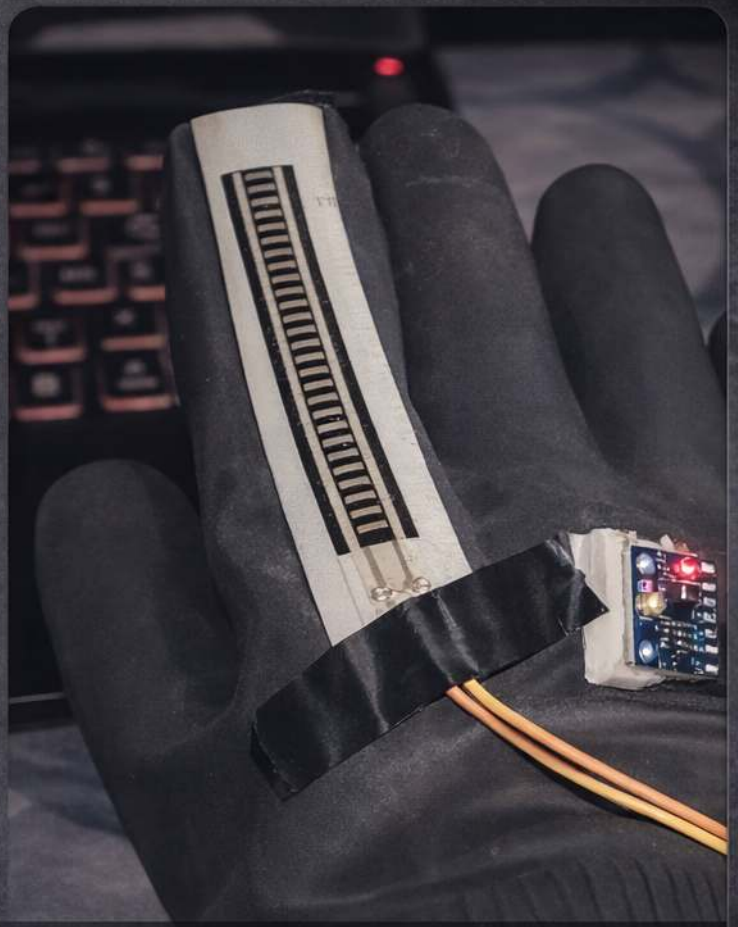
Gesture input → embedded processing → simulated pick & place

Problem

- Traditional teach-pendants and fixed programs are unintuitive.
- Need: natural, human-friendly control for flexible pick-and-place.

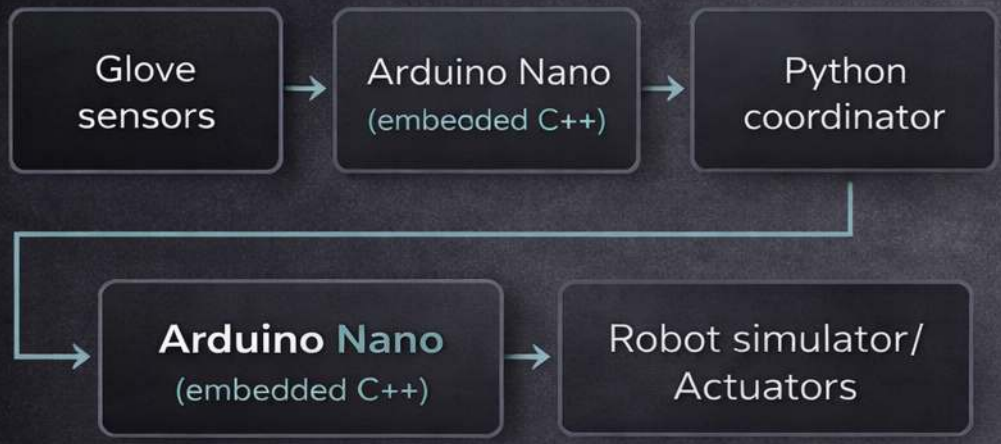
Goal:

Enable **direct hand gestures** to command a **robotic arm.**

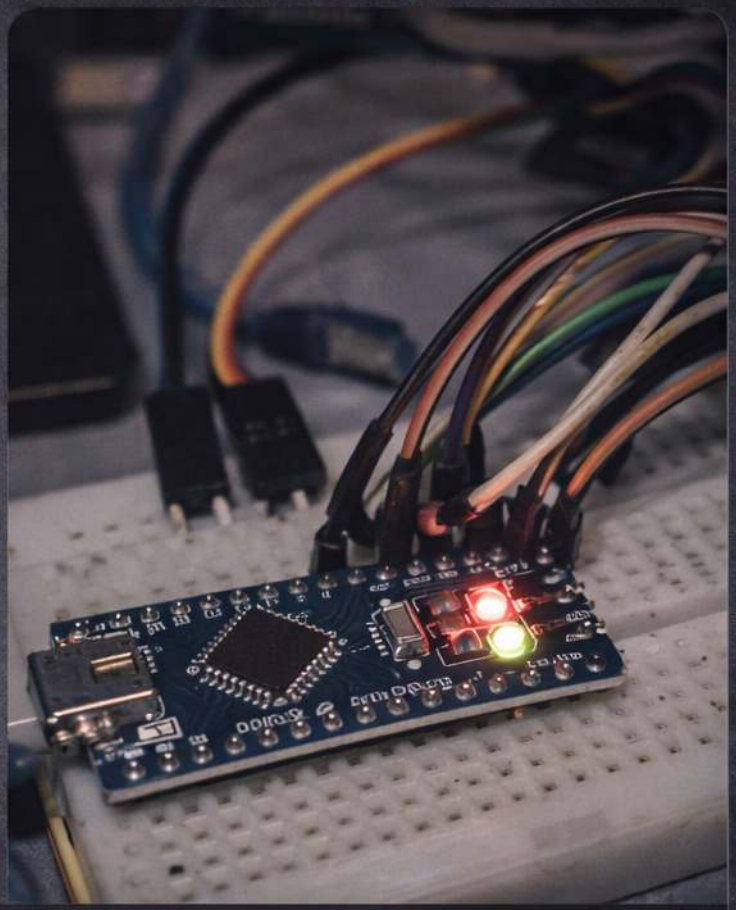


Glove + **flex sensor** captures finger motion

System architecture



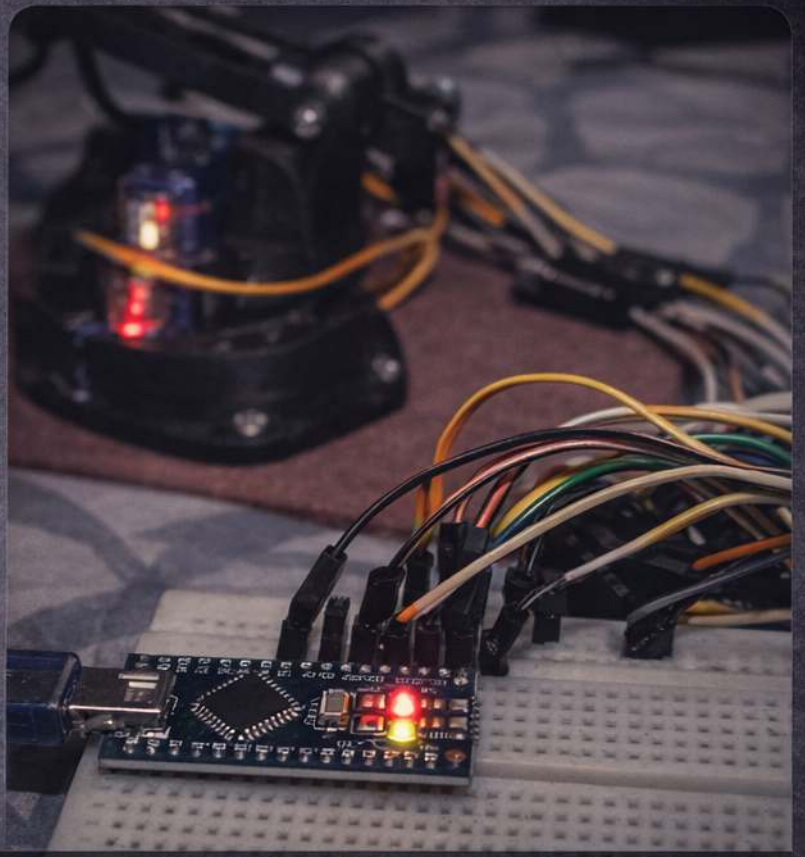
Data path is: analog sensor → ADC → mapping → servo commands (PWM)



Arduino Nano reads sensors and outputs **PWM**

Control & data flow

1. **Flex sensor / IMU** → analog readings
2. **Arduino** (C++) filters & maps to angles
3. **Python** layer manages sequences, safety checks, simulated gripper control
4. **Servo actuation** (SG90 micro servos) executes motion



Signal **processing** on device, higher logic in Python

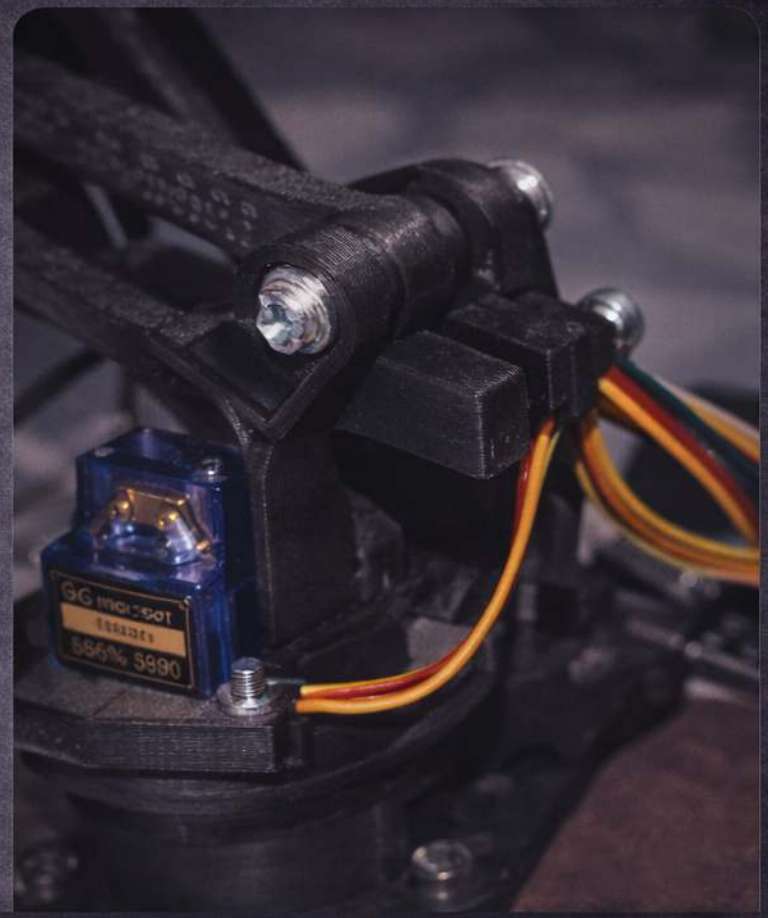
Tools & technologies

Software:

- Python – High-level control
- C++ – Embedded control logic
- Fusion 360 – Mechanical / visualization

Hardware / Concepts:

- Arduino Nano (embedded controller)
- Flex sensor (wearable)
- TowerPro SG90 micro servos ×4
- Breadboard & jumper wires, power leads / connectors



Signal **processing** on device, higher logic in Pyhsn

What I personally worked on

- Embedded firmware (sensor read, filtering, PWM mapping)
- Gesture–joint mapping and calibration
- Hardware integration (servo mounting, wiring)
- System debugging and timing checks



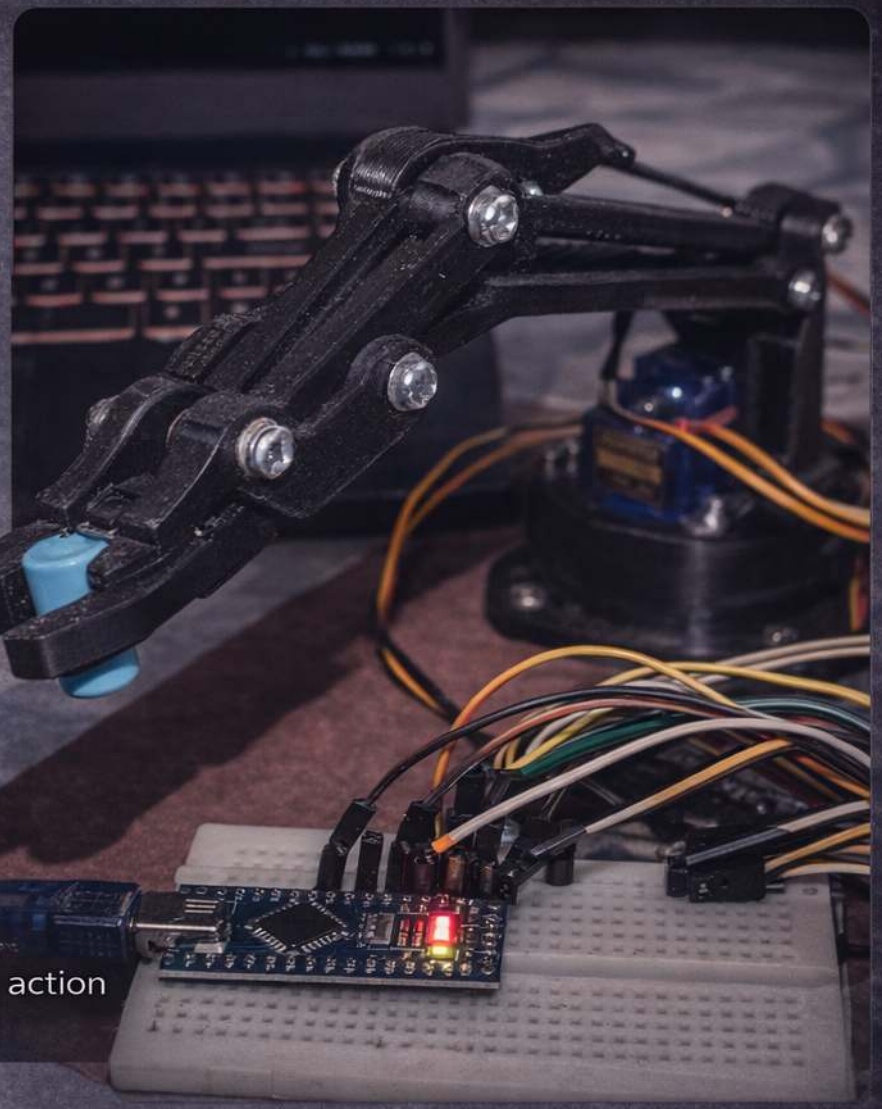
Integrated hardware, firmware and control logic

Demo highlights

- Pick & place sequence triggered reliably from gestures.
- **Latency** measured ~120–150 ms
- Gripper opens / closes, and arm moves across 3 DOF.



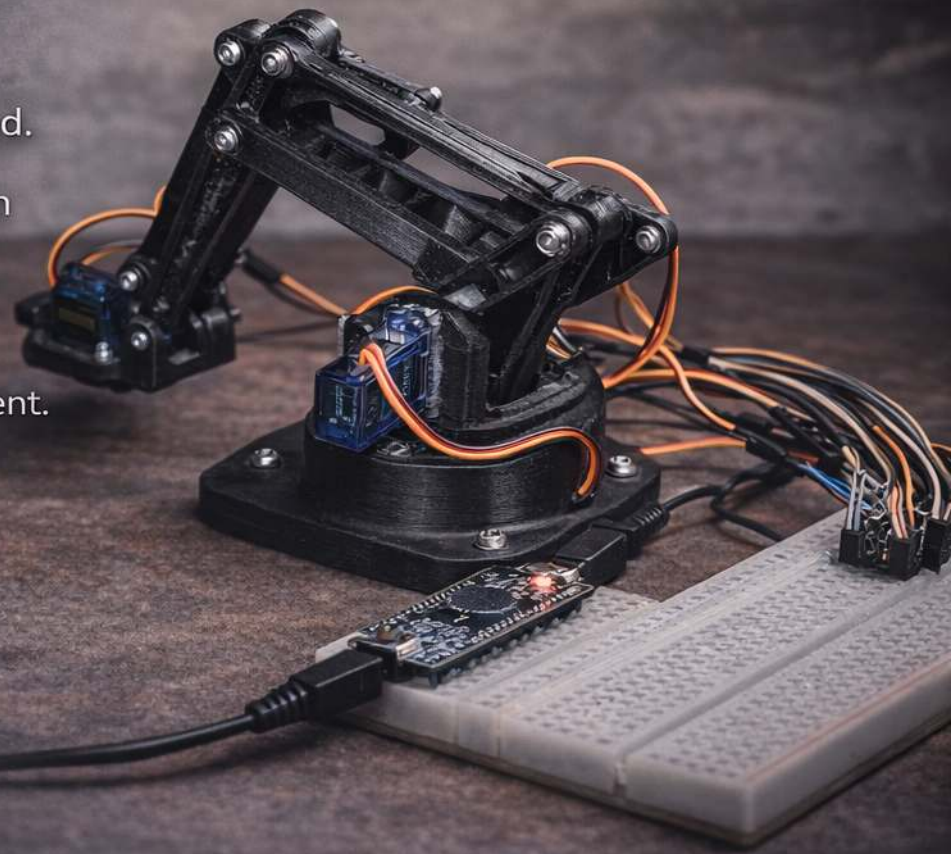
Gesture → single command → completed pick action



Conclusions

- **Prototype functions correctly and responds smoothly to gesture commands.**
- **Low latency, practical** responsiveness achieved.
- **Solid foundation** for further sensor integration and control architecture.

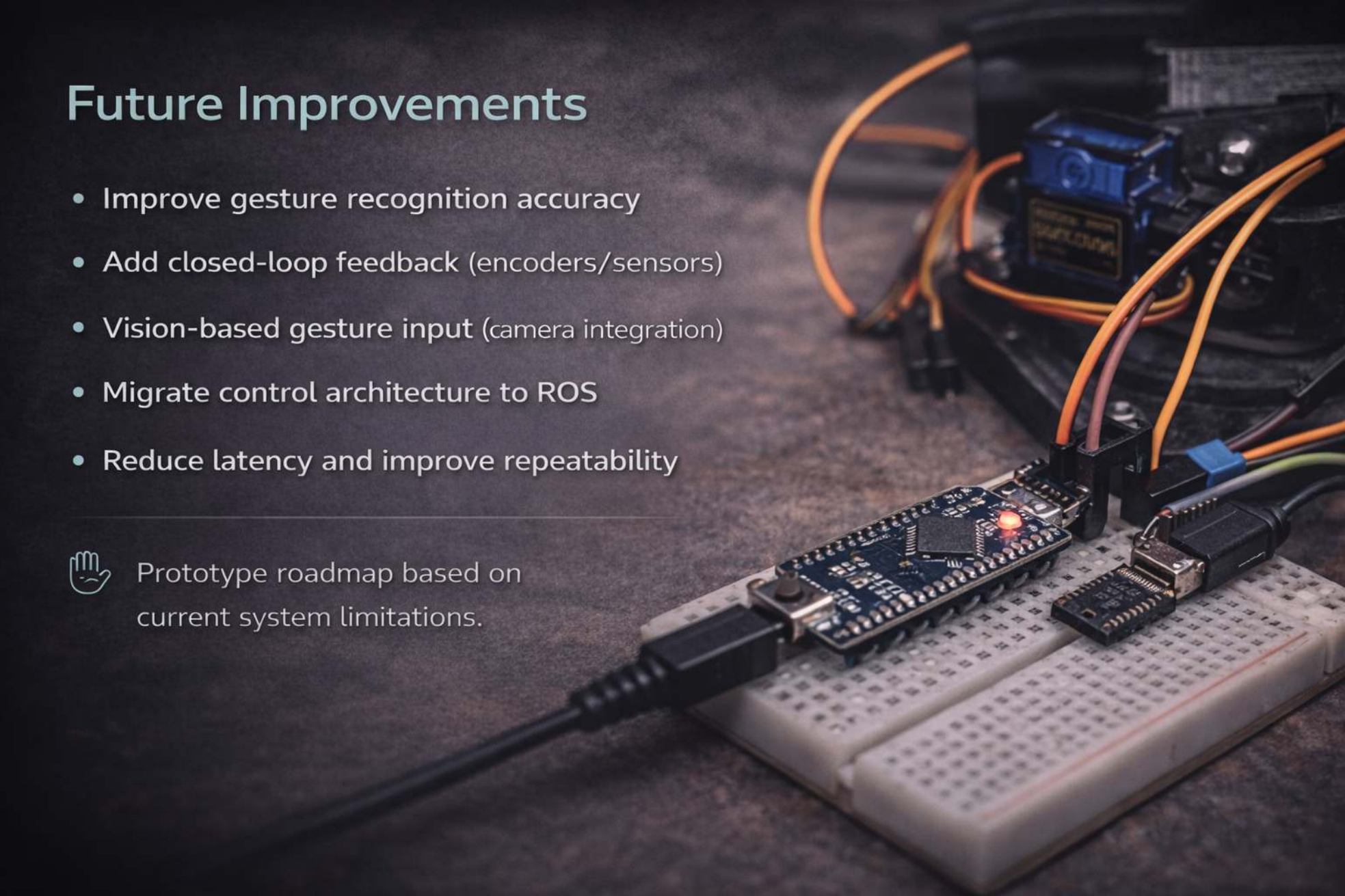
Validated prototype ready for further development.



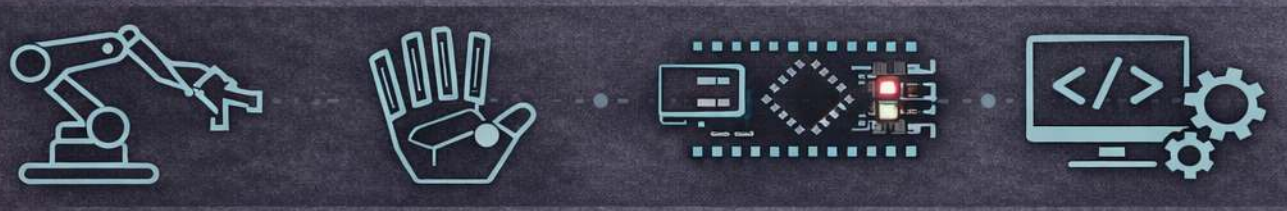
Future Improvements

- Improve gesture recognition accuracy
- Add closed-loop feedback (encoders/sensors)
- Vision-based gesture input (camera integration)
- Migrate control architecture to ROS
- Reduce latency and improve repeatability

Prototype roadmap based on current system limitations.



Thank You!



Integrated hardware, firmware and control logic